

MA1513 Weekly Self Practice Set 2

1. (a) Let $\mathbf{A} = \begin{pmatrix} 1 & a & 0 \\ 0 & 0 & 1 \\ b & 0 & 0 \end{pmatrix}$. Here a and b represent some real numbers that are not zero. Find $\mathbf{A}^{-1}$ using Gauss-Jordan elimination. (Show your working.)
- (b) Use the expansion method to find the determinant of the matrix $\mathbf{A}$ in (a). (Show your working.)
2. (a) Convert the vector equation

$$(1, -4, 4, 1) = a(1, -1, 2, 0) + b(1, 2, 0, -1)$$

into a linear system with variables a and b and write down its augmented matrix.

- (b) Use the `rref` command in MATLAB to find the value of a and b . (Do not perform Gaussian elimination.)
- (c) Use the above result to determine whether $(1, -4, 4, 1)$ belongs to

$$\text{span}\{(1, -1, 2, 0), (1, 2, 0, -1)\}?$$

Give a brief explanation.

3. Consider the matrices $\mathbf{C} = \begin{pmatrix} 12 & 34 & 56 \\ 78 & 90 & 12 \\ 34 & 56 & 78 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} 12 & 13 & 14 \\ 22 & 23 & 24 \\ 32 & 33 & 34 \end{pmatrix}$

- (a) Use MATLAB command to find the ranks of $\mathbf{C}$ and $\mathbf{D}$.
- (b) Which of $\mathbf{C}$ and $\mathbf{D}$ is full rank?
- (c) Without further working, determine which of the homogeneous systems $\mathbf{C}\mathbf{x} = \mathbf{0}$ and $\mathbf{D}\mathbf{x} = \mathbf{0}$ has non-trivial solutions.